

Tharinda Abeywardhana

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🌐 Portfolio

🌐 LinkedIn

🐙 GitHub

I like understanding why things work the way they do, then building them to find out if I'm right. Most of my projects start with a question, why does this converter ring at that frequency, why is this signal noisier than it should be, and end with a PCB and a scope probe. I've spent the last couple of years working across power electronics, embedded systems, and high-speed PCB design, and what I enjoy most is the point where the schematic meets real hardware and you have to figure out what you got wrong. I'm looking for a hardware research internship where I can work on real problems alongside people who care about getting the details right.

EDUCATION

University of Moratuwa, Sri Lanka

March 2024 – Present

- – B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering
 - **Cumulative GPA (end of 3rd Semester):** 3.72 / 4.00 (93%) [Transcript]
 - Dean's List: Semester 1

Dharmapala Vidyalaya, Colombo, Sri Lanka

- – GCE Advanced Level (2022): Physical Science Stream
 - Combined Mathematics (A*), Physics (A*), Chemistry (A*)
 - Z-Score: 2.35, Ranked 127th in Colombo District, 530th nationally (top 1.2% of 45,000+ candidates)
 - GCE Ordinary Level (2020): 9A* passes

Coursera

• University of Colorado Boulder

- Introduction to FPGA Design for Embedded Systems

Instructor: Timothy Scherr, Senior Instructor and Professor of Engineering Practice

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- Hardware Description Languages for FPGA Design

Instructor: Timothy Scherr, Senior Instructor and Professor of Engineering Practice

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- Power Electronics Specialization

Instructor: Professor Robert Erickson

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- * Introduction to Power Electronics

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- * Converter Circuits

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- * Converter Control

[View Certificate](#)

- * Magnetics for Power Electronic Converters

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• The Hong Kong University of Science and Technology

- Differential Equations for Engineers

Instructor: Professor Jeffrey R. Chasnov

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Department of Electronic and Telecommunication Engineering, University of Moratuwa

- System Verilog for ASIC/FPGA Design & Simulation

TECHNICAL STRENGTHS

Circuit Simulation & Analysis	LTspice, NI Multisim, Altair PSIM, MATLAB
PCB Design & Layout	Altium Designer (multi-layer, high-speed, power distribution)
Hardware Description Languages	SystemVerilog, Verilog, VHDL
FPGA & Digital Design	Xilinx Vivado, Intel Quartus, Verilator
Embedded Systems	STM32, ESP32, Raspberry Pi, ROS 2
Programming Languages	C/C++ (embedded), Python, MATLAB, Java
RF & Signal Processing	GNU Radio, bladeRF, MATLAB Signal Processing
Mechanical Design	SOLIDWORKS

HARDWARE PROJECTS

Smart Autonomous Floor Mopping Robot

Feb 2026 – Present

- Serving as system architect, power electronics lead, and autonomous navigation engineer within a team of five building a fully autonomous floor cleaning robot.
- Designed a power management PCB with LiPo charging/discharging control and three regulated rails (3.3 V, 5 V, 12 V) with per-rail current monitoring, budgeted for a 3+ hour continuous runtime across three subsystems.
- Designed a sensor management PCB with an STM32 microcontroller for cliff and obstacle detection, wheel encoder interfacing, pump control, and closed-loop motion feedback.
- Leading the autonomous navigation stack using LiDAR-based SLAM, path planning, and coverage navigation on Raspberry Pi with ROS 2.
- Integrating the power, sensing, and navigation subsystems with real-time power budgeting and thermal management.

Tools: Altium Designer, STM32, LTspice, ROS 2, LiDAR, Raspberry Pi

Status: In Development

Repository:  GitHub

Advanced 12-Layer High-Speed PCB System Design - iMX6 Rex

Jan 2026 – May 2026

- Designed a 12-layer PCB from scratch based on the open-source FEDEVEL iMX6 Rex schematic (NXP i.MX6, up to 1.2 GHz quad-core, DDR3-1066).
- Engineered a 12-layer stackup (Signal/GND/Signal/GND/Power×4/GND/Signal/GND/Signal) optimized for signal integrity and low-impedance power delivery.
- Routed length-matched, impedance-controlled differential pairs for DDR3, PCIe, SATA, Gigabit Ethernet, HDMI, LVDS, and USB with controlled return paths.
- Performed BGA fanout/escape routing and designed a multi-rail PDN (1V375, 1V5, 2V5, 3V3) with hierarchical decoupling to minimize voltage droop.
- Applied EMI/EMC reduction techniques, thermal-aware component placement, and DFM/DRC validation in Altium Designer (completed through design and verification).

Tools: Altium Designer

Repository:  GitHub

Buck Converter Design - MIT 6.622 Specification

March 2026 – May 2026

- Designed a high-efficiency synchronous buck converter (24–36 V input, 14 V output, 50–125 W) to the MIT 6.622 specification, switching at 100 kHz.
- Designed a Type III analog compensator in MATLAB achieving a **+50.96° phase margin** and infinite gain margin, after analytically ruling out Type I and Type II for the LC-resonant plant ($f_{LC} \approx 1.79$ kHz).
- Verified closed-loop regulation in Altair PSIM across a 20–40 V input sweep, holding output at 13.84 V with ripple under 100 mV.
- Achieved **97.69% simulated efficiency** at 28 V, exceeding the 90% requirement by about 8 points, supported by a full loss budget down to 3.5% total losses.
- Confirmed thermal safety (MOSFET $\approx 76^\circ\text{C}$, diode $\approx 104^\circ\text{C}$ junction) across -20 to $+50^\circ\text{C}$ ambient through device-level thermal modeling.

Tools: LTspice, MATLAB, Altair PSIM, Altium Designer

Repository:  GitHub

Smart Inverter Design and Control System

May 2026 – Present

- Designing a high-efficiency DC-AC inverter targeting 2 kW output capability and 24 V DC to 230 V AC conversion.
- Implementing closed-loop sinusoidal PWM control with real-time feedback for output voltage regulation and harmonic suppression.
- Developing multi-stage LC filtering to achieve low total harmonic distortion and compliance with grid standards.
- Integrating protection circuits for overcurrent, overvoltage, and thermal management with fault detection.
- Working on the gate driver and control PCB design for high-speed switching frequency operation and isolated MOSFET/IGBT drive signals.

ThermoBand - Wearable Temperature Monitoring Device

Feb 2025 – Jul 2025

- Designed two custom 4-layer PCBs for a wearable temperature monitoring system with wireless telemetry.
- Developed the power management PCB with LiPo battery charging/discharging control, integrated voltage regulation, multi-rail power distribution, and protection circuitry, achieving **12+ hours** of continuous battery operation.
- Designed the sensor interface PCB with precision analog conditioning (precision amplification and low-pass filtering), reaching $\pm 0.01^\circ\text{C}$ measurement resolution.
- Integrated on-board ESP32 wireless telemetry and optimized the signal path for noise rejection within a wearable form factor.

Tools: Altium Designer, Solidworks

Repository:  GitHub

Smart IV Fluid Monitoring and Alert System

Jul 2025 – Dec 2025

- Designed a custom 2-layer PCB integrating an ESP32 microcontroller, IR sensing front-end, and Wi-Fi telemetry for real-time IV fluid monitoring.
- Implemented an analog sensor conditioning stage for IR signal processing with amplification and filtering, resolving drip rates down to **1 drop per minute** for accurate low-flow detection.
- Developed the system for continuous monitoring with real-time wireless alerting on low-flow and empty-bag conditions.
- Managed power delivery constraints and battery efficiency to support reliable continuous operation.

Tools: Altium Designer, ESP32 Firmware, Solidworks

Repository:  GitHub

Fully Analog Acoustic Guitar Tuner

Jul 2025 – Dec 2025

- Designed a fully analog guitar tuning device with no digital signal processing.
- Developed the analog signal-conditioning chain including a low-noise microphone preamplifier and precision band-pass filters covering all six strings at their fundamental frequencies.
- Implemented a frequency-to-voltage conversion stage for tuning indication, achieving **± 2 Hz accuracy**, validated by tuning a real guitar.
- Designed a shielded enclosure with proper grounding and EMI mitigation to preserve front-end signal integrity.

Tools: LTspice, Altium Designer, Solidworks

Repository:  GitHub

ADDITIONAL ENGINEERING PROJECTS

Two-Way Digital Paging System - Software Defined Radio

Jul 2025 – Dec 2025

- Designed and implemented a complete two-way communication system using SDR hardware (*bladeRF*) instead of fixed radio transceivers.
- Built the full baseband signal processing chain including audio conditioning, packet framing with addressing, acknowledgment handling, and digital modulation using BPSK/QPSK.
- Deployed the system on the GNU Radio framework for end-to-end implementation and testing.
- Validated system operation through real-world RF experiments.

Tools: GNU Radio, bladeRF, MATLAB

Repository:  GitHub

VOLUNTEER AND LEADERSHIP EXPERIENCE

Electronic and Telecommunication Engineering E-Club, University of Moratuwa

- **HR Coordinator** Aug 2025 – Present
- **Event Organizer (Eid26)** Aug 2025 – Present

Sasnaka Sansada Foundation

- **LearnSteer National Project Coordinator** Dec 2024 – Nov 2025
Led nationwide Physics workshops and examinations for GCE Advanced Level students.
- **Member, Colombo District Team** Jun 2023 – Present
Educational outreach and teaching support for rural school students.

REFERENCES

Dr. Ranga Rodrigo

Senior Lecturer

Department of Electronic and Telecommunication Engineering

University of Moratuwa, Sri Lanka

B.Sc.Eng.Hons. (Moratuwa, Sri Lanka), M.E.Sc. (Western, Canada), Ph.D. (Western, Canada)

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Dr. Upeka Premaratne

Senior Lecturer

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